

Healthcare Sales Operations

*Referral Performance &
Workflow Improvement*

PROJECT BRIEF AND MEASUREMENT CONTRACT

SQL | Power BI | Operational improvement playbook

Project owner	Scenario and observation period
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Starter version 0.1 Prepared September 30, 2026	Receipts: January 1, 2025-August 31, 2026 Frozen snapshot: August 31, 2026

The decision

Where are referrals getting delayed or failing to progress, which provider accounts need attention, and what should sales and intake operations change?

The operating scenario

A fictional multisite outpatient organization works with 120 provider accounts across six territories and eight sites. Twelve sales representatives manage partner relationships; eight intake owners manage referral progression. Those responsibilities are modeled separately.

The first demonstrable problem

The export contains 6,300 rows for 6,000 referral IDs. Counting rows would overstate referral volume by 5.0% relative to the unique-referral baseline. The SQL preserves the raw records, counts one referral per ID, and documents every repeated ID.

What this starter version includes

Synthetic source data, a reproducible SQLite workflow, reporting CSVs, 40 passing automated data/SQL checks, a requirements and KPI dictionary, draft workflow instructions, and a Power BI build guide. The Power BI report, manual UAT, and any operational pilot are not yet complete.

All organization, account, referral and employee data is synthetic. No employer records or actual patient data was used. AI assisted the starter code and documentation; the project owner should review, adapt and explain the work before presenting a completed case study.

Requirements before visuals

The report serves three different decisions. These are proposed scenario requirements, not claims that employer stakeholders have been interviewed or approved them.

Page	Primary decision	Accountable user
Growth & Referral Performance	Distinguish volume, cohort booking, completed first visits and territory targets.	Growth leader
Territory & Partner Accounts	Select a partner action after checking intake backlog, account mix and outreach.	Sales manager
Workflow & Data Quality	Assign owners, review uncertain data, and create a dated next action.	Intake supervisor / data steward

Required behaviors

Requirement	Acceptance expectation
One counted referral	One canonical row per referral ID; all original versions remain available.
Defensible conversion	Only timing-valid referrals with 30 full elapsed days enter the same numerator/denominator cohort.
Date-role clarity	Receipt-month cohorts and completion-month activity are separate; neither is presented as the other.
Actionable ownership	An open referral without intake ownership appears in a review queue; a sales rep is not substituted.
Visible uncertainty	Missing dates, ambiguous statuses and conflicting records remain review items, not guessed corrections.
Target-grain integrity	Targets are compared only at month/territory level; unsupported filters return a blank target.
Snapshot integrity	Current backlog uses August 31, 2026; receipt-date slicers do not silently hide old unresolved work.

Keep the scope focused

One referral-to-first-appointment process; three report pages. No actual medical decision-making, patient identities, repeated treatment visits, rescheduling, real CRM integration, revenue forecasting, geographic market maps, or causal claims about representatives.

Full requirement IDs and their SQL/manual-test links appear in docs/requirements.md. Manual acceptance cases start as Not run in tests/manual_uat.csv.

The measurement contract

Every interval uses date-only calendar days. The snapshot is fixed at August 31, 2026. The two-day follow-up and seven-day stalled-stage thresholds are project assumptions, not regulatory or clinical deadlines.

Measure	Definition / interpretation
Dated referrals	Count canonical referrals with a valid receipt date. The 36 missing-receipt records remain in the audit population.
Completed first visits	Count timing-valid first completed visits by completion date. This is activity, not referral conversion.
Mature 30-day cohort	Valid receipt and stage dates; receipt age at least 30 days. Closed and unscheduled referrals remain in the cohort.
Scheduled within 30 days	First booking from day 0 through day 30, inclusive, within that same mature cohort. Booking is not attendance.
Median days to booking	Only valid, observed bookings. Pair with unresolved-stage age because unfinished transitions are excluded.
Scheduling work queue	Current stage is Received, In Review, or Ready to Schedule. Already Scheduled cases are not in this pre-booking queue.
Overdue follow-up	Pre-booking case with an open task due before the snapshot. Due today is not overdue; missing due dates are separate.
Target attainment	Dated receipts divided by the corresponding month-territory target. No account, site or payer allocation is assumed.

The 30-day boundary matters

An August 1 receipt is mature on August 31; an August 2 receipt is not. A July 1 receipt booked July 31 qualifies; a booking on August 1 does not. An immature referral is excluded even if it has already been booked.

Baseline reconciliation

5,602 mature referrals + 320 not yet mature + 78 timing-excluded referrals = 6,000 total. Of the mature referrals, 4,024 were booked within 30 days: 71.8%. All figures describe generated data, not external performance benchmarks.

Only August 1 receipts are mature within the August receipt cohort. Use January 2025-July 2026 for complete-month cohort comparisons, while retaining August in incoming-volume and completed-visit activity reporting.

Data design and SQL controls

The source data includes accounts, territories, sites, sales representatives, intake owners, referral exports, stage events, follow-up tasks, provider outreach, targets and status mappings.

Reporting layer	Grain	Rows
fact_referrals	One canonical referral	6,000
fact_activities	One provider outreach activity	3,600
fact_targets	One month and territory	120
fact_account_snapshot	One account at fixed snapshot	120
fact_exceptions	One entity and exception rule	784
Dimensions / configuration	Dates, accounts, territories, sites, owners, snapshot	Separate tables

The SQL workflow

Profile source counts and statuses. Rank export rows by source update, ingestion time and source row ID. Retain one survivor per referral. Aggregate stage events and tasks before joining them to referrals. Build timing/cohort/queue flags. Export separate facts and dimensions for Power BI.

This avoids multiplying referral rows when joining one-to-many task and stage histories. Targets and outreach remain separate facts rather than being repeated on each referral row.

Intentional quality cases

Case	Treatment
300 excess export rows	Count one referral; preserve 260 repeat exports and 40 older versions in the raw data.
36 missing receipt dates	Retain referrals for review; do not place them into a receipt-month chart.
24 backdated / 18 missing stage dates	Exclude those referrals from timed metrics and show the exclusion count.
18 unknown / 24 conflicting statuses	Use event sequence provisionally; retain the mapping/conflict review item.
150 missing intake / 6 missing sales owners	Preserve separate unassigned members; do not invent an owner.

A referral can have multiple exceptions, so 784 issue records does not mean 784 different referrals. Repeated-ID issues are resolved in reporting; other source corrections require review. Full field definitions are in docs/data_dictionary.csv.

From exception to owned action

The proposed improvement is a daily exception-review process with a named accountable role and closure evidence. It has not been deployed or shown to improve real-world outcomes.

1. Verify the snapshot

The data steward reconciles raw counts, unique IDs, date scope and test results before publication. A structural failure stops a new report release; preserve the last validated snapshot with a freshness warning.

2. Separate owner assignment from record correction

The intake supervisor reviews open referrals without an intake owner. Sales operations assigns missing provider-account sales owners. The data steward and intake supervisor verify ambiguous dates or statuses instead of guessing values.

3. Review follow-up and coordinate outreach

Intake owners review overdue tasks, missing due dates, and pending cases without an open task as separate categories. Sales representatives address partner communication after checking intake constraints; more referral volume is not automatically the right first action.

4. Close the issue with evidence

Capture the original value, corrected value, rationale, owner, change date and reviewer. Rebuild the reporting layer and confirm that the issue clears without changing unrelated referral counts. Source records are not overwritten automatically.

Accountable role	Owned control
Data steward	Count integrity, version selection, status mappings and regression tests.
Intake supervisor	Case-owner assignment, uncertain referral dates/statuses and queue oversight.
Intake owner	Dated next action, follow-up completion and documented escalation.
Sales operations / sales manager	Partner-account ownership and appropriate outreach after operational review.

What success would be measured against

The synthetic snapshot has 737 pending scheduling referrals, 408 with overdue follow-up, and 21 open referrals without intake owners. The groups overlap. A proposed pilot would track these measures with clearly defined denominators and incoming volume; it would not treat this generated baseline as actual operating performance.

No appointment uplift, hours saved, financial return or clinical outcome has been demonstrated. The full procedure is in docs/workflow_playbook.md; the leadership memo is explicitly a starter draft.

Validation and the first build milestone

Ready in this package	Still to perform
Synthetic source data and a fixed-seed generator.	Review and adapt the scenario in the project owner's own words.
SQLite database, visible queries, and Power BI import CSVs.	Create the Power BI model, relationships and measures.
40 passing automated SQL/data checks with saved evidence.	Complete 12 manual acceptance cases in Power BI.
Requirements, definitions, draft SOP and leadership memo.	Build the three pages, export the actual report preview and finalize the recommendation.

Your first SQL exercise

Open database/healthcare_sales_operations.sqlite in a SQLite client. Run the first query in sql/01_profile_sources.sql. Explain why the results are 6,300 source rows, 6,000 unique referrals and 300 excess rows. Then review the ROW_NUMBER survivor rule in sql/02_build_reporting.sql.

Your first Power BI milestone

Use the Text/CSV import workflow for the seven core tables listed in powerbi/BUILD_GUIDE.md. Add the five starter measures before styling any page. With no filters, reconcile to 6,000 unique referrals, 5,964 dated referrals, 5,602 mature referrals, 4,024 booked within 30 days, and 71.8%.

The supplied DAX has not been executed in Power BI. SQL/data tests are not human UAT. Test completion-date relationships, the August observation boundary, target-filter behavior, and snapshot queues before calling the report complete.

Reference notes

The employer postings informed skill themes only. They are not sources of the synthetic data or operational targets. Technical guidance supports the implementation approach. Accessed September 30, 2026.

IVX Health - Manager, Sales Operations
<https://job-boards.greenhouse.io/ivxhealth/jobs/4290169009>

SQLite - window functions
<https://www.sqlite.org/windowfunctions.html>

Microsoft - Power BI star-schema guidance
<https://learn.microsoft.com/en-us/power-bi/guidance/star-schema>

Microsoft - Text/CSV connector
<https://learn.microsoft.com/en-us/power-query/connectors/text-csv>

Additional Cognizant, DAX and public-sharing references are in README.md and powerbi/BUILD_GUIDE.md. No GitHub changes or public report publishing has been performed.